Editors' Suggestion

Fluxoid Solitons in Superconducting Tapered Tubes and Bottlenecks

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A thin-walled tubular superconductor develops a quantized fluxoid in the presence of an axial magnetic field. The fluxoid corresponds to the number of phase windings of the superconducting order parameter and is topological in nature. When the tube has a radius variation along the axial direction, forming a bottleneck structure between sections with different radius, a fluxoid mismatch can appear depending on the applied magnetic field. The bottleneck then becomes a topological boundary and is host to topologically protected solutions for the order parameter, dubbed fluxoid solitons, that are free to move around bottlenecks with cylindrical symmetry. Fluxoid solitons are a new type of vortex with nonquantized flux, loosely related to Pearl vortices in thin superconducting films, and fluxons in Corbino Josephson junctions. We characterize their properties as a function of system parameters using the self-consistent quasiclassical theory of diffusive superconductors. We consider both short bottleneck structures and long tapered tubes, where multiple trapped fluxoid solitons adopt elaborate arrangements dictated by their mutual repulsion.

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Introduction—The study of thin-walled tubular superconductors in an axial magnetic field dates back to the pioneering experiments by Little and Parks [1,2], and the subsequent analysis by Tinkham [3] in the framework of the Ginzburg-Landau theory [4]. These seminal works demonstrated that, in a multiply connected superconductor such as, e.g., a superconducting tube, the phase of the complex order parameter $\Delta(\boldsymbol{r})$ can develop discrete windings around the tube in response to a longitudinal magnetic field [5], a result already anticipated by London [6]. The winding number n, also known as *fluxoid* number [3,7,8], is quantized, even if the magnetic flux Φ into the tube is not (for thin-walled tubes). Specifically,

$$n = \lfloor \Phi/\Phi_0 \rceil, \tag{1}$$

where $\lfloor x \rceil$ denotes the integer closest to x and $\Phi_0 = h/2e$ is the superconducting flux quantum (h being Planck's constant and e the elementary charge).

Recent years have seen renewed interest in the study of thin superconducting tubes as a result of the successful growth and characterization of so-called full-shell nanowires [9]. These hybrid nanowires consist of a semiconductor core fully surrounded by a thin s-wave superconductor shell and subjected to an axial magnetic field [10], and have been explored as possible candidates to realize topological superconductivity and Majorana zero modes [11–16]. Theoretical modeling [10,17–26] and experiments [10,27–35] have demonstrated that fluxoid quantization in the shell (together with the semiconducting core) plays a key role in the transport and spectral phenomenology of these nanowires, including the appearance of analogs of Caroli–de Gennes subgap states [22,35], a nonconventional Josephson effect [23,31,32,36–39], the mechanism behind topological superconductivity [10,17,18,24,26], and a fluxoid-valve effect [40], to name a few.

Focusing on the properties of a thin superconducting tube, and regardless of the material in the core, the significance of fluxoid quantization is twofold. First, jumps in the fluxoid number affect the superconducting properties of the nanostructure, such as critical temperature, superconducting gap, and equilibrium supercurrents, which then oscillate with field in a nontrivial reemergent manner [the so-called Little-Parks (LP) effect [1–3]]. Second, the fluxoid quantization has a topological nature. Each field-induced jump in the fluxoid involves a topological transition between two fundamentally incompatible configurations of the order parameter. As in other topological phases, such as topological insulators and superconductors [41], bringing together two tubes with different topology of their order parameter should give rise to topologically protected solutions at the boundary that depend on the difference in topological invariants in the bulk (the so-called bulk-boundary correspondence principle [41–44]).

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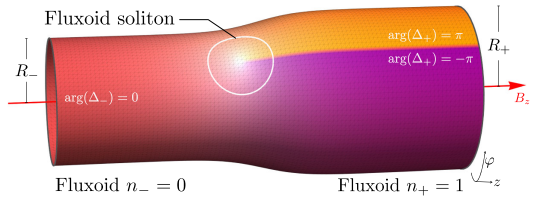

FIG. 1. Sketch of a superconducting tubular bottleneck. A thin-walled diffusive superconducting tube has a variation of its radius along the axial direction z. A geometrical defect or bottleneck is created between sections of different radii R_- and R_+. In the presence of a longitudinal magnetic field B_z, the phase of the superconductor order parameter $\Delta(\boldsymbol{r})$ acquires a radius-dependent integer number n of windings, called fluxoids, so $\arg(\Delta) = n\varphi \bmod 2\pi$. At the bottleneck, the fluxoid is forced to change abruptly from n_- to n_+, giving rise to $|\delta n| = |n_+ - n_-|$ fluxoid solitons in $\Delta(\boldsymbol{r})$ that are free to move along the azimuthal direction φ for cylindrically symmetric systems.

In this Letter, we demonstrate the emergence of solitonic solutions at fluxoid boundaries and analyze their properties. A fluxoid boundary can be induced in a thin superconducting tube under a uniform magnetic field by making its radius position dependent, so that it transitions continuously from a smaller radius R_- to a larger one R_+ through a bottleneck region; see Fig. 1. To our knowledge, the detailed properties of this kind of superconducting structure have not yet been analyzed in the literature, despite recently becoming an experimental reality [45] thanks to advances in full-shell nanowire growth techniques. We study this system using a fully self-consistent quasiclassical theory of diffusive superconductors [46] that can deal with systems of any geometry at arbitrary temperature, unlike the conventional Ginzburg-Landau formalism. Using this approach, we demonstrate that a fluxoid mismatch between the two sides of the bottleneck gives rise to a peculiar kind of order parameter defect, dubbed *fluxoid soliton*. Fluxoid solitons are topologically protected nodes of the order parameter surrounded by a vortex of circulating supercurrent density and accompanied by a nonquantized magnetic flux. Fluxoid solitons are related to Pearl vortices in thin films [47,48], to solitons in two-band superconductors [49–53], and to fluxons trapped in long Josephson junctions [54–60], specifically with Corbino geometries [61–64]. As expected from topological boundary modes, fluxoid solitons are pinned to the bottleneck region, but for cylindrically symmetric tubes, they can move freely around it (although their motion may still be damped [55,61]). Like Pearl [47] and Abrikosov [65] vortices, fluxoid solitons repel each other. Hence, as the bottleneck length grows, the solitons transition between a necklacelike arrangement around the tube in short bottlenecks to complex longitudinal configurations in bottlenecks significantly longer than the superconducting coherence length ξ_0 (the so-called "tapered" geometry, where the tube radius changes slowly along the axial direction).

Model and methods—We consider a thin hollow cylinder at temperature T made of a diffusive single-band, s-wave superconducting material such as aluminum, with a zero-temperature superconducting coherence length ξ_0, a bulk critical temperature T_C^0, and a zero-temperature bulk pairing $\Delta_0 \approx 1.764 k_B T_C^0$ [3]. We assume that the radius of the thin-walled tube $R(z)$ varies along the axial direction, as shown in Fig. 1, so that $R(z \to \pm\infty) = R_\pm$. This geometrical variation is concentrated in a region that we call the bottleneck, of length L_b. We distinguish between short bottlenecks with $L_b \lesssim \xi_0$, modeled with a $\tanh(z/L_b)$ profile for $R(z)$, and long bottlenecks with $L_b \gg \xi_0$, dubbed tapered tubes and modeled with a linear $R(z)$ profile. An axial magnetic field $\boldsymbol{B} = B_z\hat{\boldsymbol{z}}$ is applied to the tube [66], so that the magnetic flux at each z section reads $\Phi(z) = \pi R(z)^2 B_z$, with $\Phi(z \to \pm\infty) = \Phi_\pm$. This defines a local fluxoid number $n(z) = \lfloor \Phi(z)/\Phi_0 \rceil \in \mathbb{Z}$, with asymptotic values $n(z \to \pm\infty) = n_\pm$. A "fluxoid mismatch" is defined as a nonzero integer $\delta n = n_+ - n_-$. The tube wall thickness is considered to be much smaller than the London penetration length λ_L, so that the magnetic field is assumed unscreened and spatially uniform [67]. The Zeeman effect is neglected.

Our goal is to compute the self-consistent complex order parameter $\Delta(z, \varphi)$ in a zero-thickness tube, where z, φ are its cylindrical coordinates. To this end, we apply the nonuniform, arbitrary-temperature, self-consistent Usadel theory in $d = 2$ dimensions, summarized in Supplemental Material [68]. It involves the numerical minimization of a grand canonical functional $\Omega[\hat{g}]$ over possible quasiclassical Green's functions $\hat{g}$. For the simple case of a constant-R tube, we recover the London result $\Delta = |\Delta| e^{in\varphi}$, with n given by Eq. (1), and the LP phenomenology.

Results—In a tube with a nonuniform bottleneck $R(z)$ the London solution for Δ is recovered asymptotically, and is denoted by $\Delta_\pm = \Delta(z \to \pm\infty, \varphi) = |\Delta_\pm| e^{in_\pm \varphi}$. In the presence of a fluxoid mismatch $\delta n \neq 0$, the winding numbers $n_\pm$ of the asymptotic $\Delta_\pm$ are different, and hence the equilibrium complex solution $\Delta(z, \varphi)$ necessarily becomes z dependent around the bottleneck. $\Delta(z, \varphi)$ must be continuous and differentiable (the current density should be well defined; see Supplemental Material [68]), smoothly connecting two topologically incompatible solutions with different phase windings. As illustrated analytically in Supplemental Material [68], the only way to do so requires $\Delta(\boldsymbol{r})$ to vanish at some point around the bottleneck region. The actual solution, computed numerically by minimizing the Ω functional, confirms this.

Short bottlenecks—Figure 2 shows the $\Delta(\boldsymbol{r})$ solution for short bottlenecks $L_b \sim \xi_0$. We see that $|\Delta|$ actually vanishes at a number $|\delta n|$ of points around the bottleneck [dark regions in Figs. 2(a), 2(c), 2(e), and 2(g)], with the phase $\arg(\Delta)$ winding by $\pm 2\pi$ around each of them. Each of these zeros is the core of a fluxoid soliton. The $\arg(\Delta)$ plots show, moreover, that each soliton lies at the end of

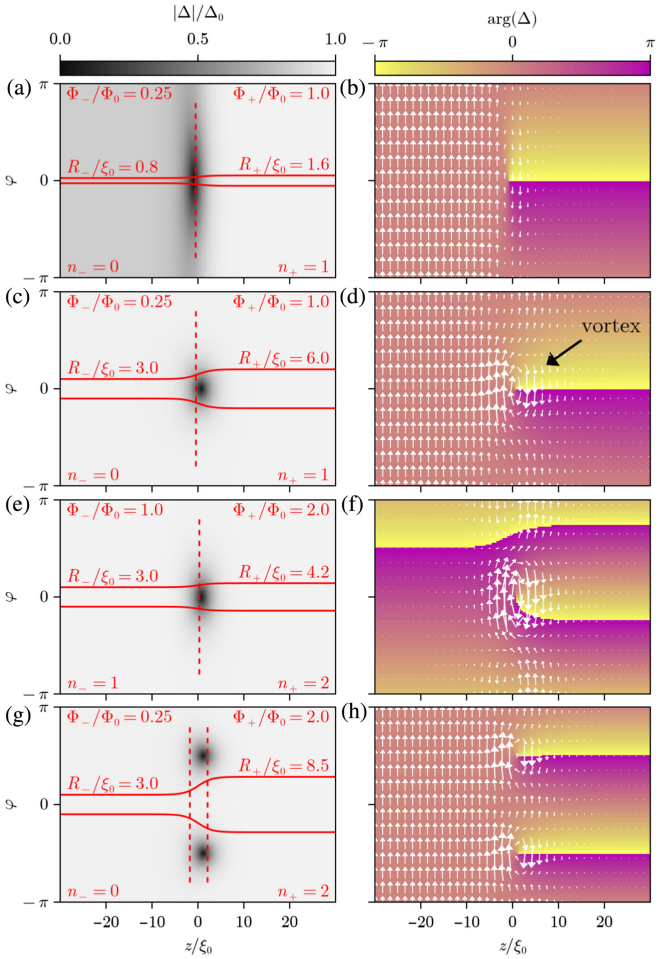

FIG. 2. Fluxoid solitons in short tubular bottlenecks. Pairing modulus $|\Delta|$ (a) and phase $\arg(\Delta)$ (b), versus cylindrical coordinates φ and z (normalized to the superconducting coherence length ξ_0), for a bottleneck of length $L_b = 3\xi_0$ represented by the solid red line in (a). Temperature is $T = 0.25T_C^0$, and B_z field is such that the dimensionless flux is $\Phi_-/\Phi_0 = 0.25$ on the R_- section, and $\Phi_+/\Phi_0 = 1$ on the R_+ section. The dashed red line shows the position at which the local fluxoid $n(z)$ jumps by one. The total fluxoid mismatch is $\delta n = 1$, so one soliton emerges at the bottleneck, close to the dashed line [black shadow in (a)]. The associated supercurrent density $\boldsymbol{J}$ [white arrows in (b)] forms a vortex around the soliton. Panels (c) and (d) are like (a) and (b) but for larger $R_\pm/\xi_0$ and the same flux. Panels (e) and (f) are similar to (c) and (d) but with an integer flux on both sides of the bottleneck, which makes currents vanish asymptotically. Panels (g) and (h) are like (c) and (d) but with $\delta n = 2$ solitons.

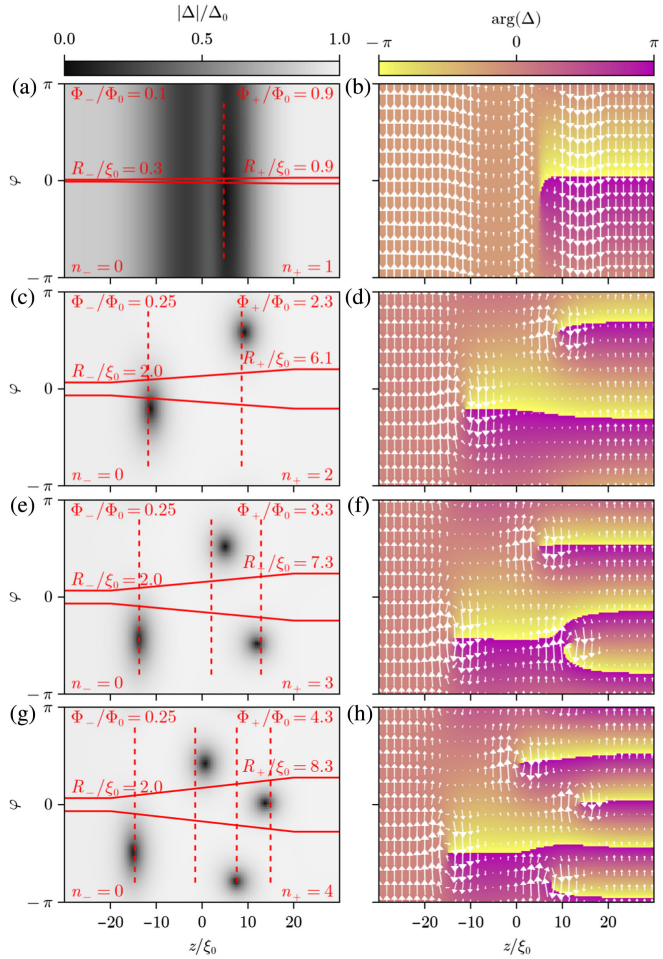

FIG. 3. Fluxoid solitons in tapered tubes. Four tapered nanowire geometries with increasing number of solitons are represented from top to bottom, with the same plotting conventions as in Fig. 2. Unlike in the short bottleneck case, solitons are spaced along the tapered section (z direction), with their equilibrium positions governed by their mutual repulsion. In the case of a narrow tube (a),(b), the destructive LP effect transforms solitons into strips of near-zero $|\Delta|$ across a finite z interval.

a "$\pm\pi$-phase string" shown as a yellow-pink color discontinuity [Figs. 2(b), 2(d), 2(f), and 2(h)]. This is a manifestation of the $\pm 2\pi$ phase winding of Δ around each soliton [94]. When the bottleneck hosts several fluxoid solitons [Figs. 2(g) and 2(h)], they distribute around the tube in a way that maximizes their distance, revealing a vortexlike repulsion between them. Since the bottleneck length in Fig. 2 is $L_b = 3\xi_0$, of the order of the soliton diameter, solitons can only separate along the φ direction, leading to a necklacelike multisoliton configuration around the bottleneck.

The equilibrium supercurrent density $\boldsymbol{J}$, shown with white arrows in Figs. 2(b), 2(d), 2(f), and 2(h), circulates around solitons. These current vortices produce a specific magnetic signature that could be used to image the solitons in real space, as discussed in Supplemental Material [68] (see also Ref. [95]). Note that at $z \to \pm\infty$ the current does not have a J_z component because no current is injected externally into the system. However, the asymptotic azimuthal component of the current J_φ is finite in general. It represents a screening response to the applied magnetic field B_z. In the particular case of commensurate flux, $\Phi_\pm/\Phi_0 \in \mathbb{Z}$, there is no screening response and the current J_φ vanishes exponentially at large $|z|$; see Fig. 2(f). A detailed characterization of the soliton profile and the

asymptotic $|\Delta_\pm|$ as a function of tube radii, flux, and temperature is included in the End Matter.

Tapered tubes—Bottleneck regions of increasing length allow for other nontrivial boundary soliton configurations. Tubes with linear $R(z)$ profiles across regions significantly longer than the typical soliton width $2\xi_0$ are dubbed here as "tapered" tubes. Narrow tapered tubes with $R_\pm \lesssim 0.6\xi_0$ [96] may enter the destructive LP regime locally within certain z intervals. This leads to a strip of suppressed pairing in those intervals. A careful analysis shows that $|\Delta|$ only vanishes exactly at a single point in each interval, so the strip is in fact a very distorted fluxoid soliton. This is shown in Figs. 3(a) and 3(b) for $\delta n = 1$ (single destructive strip). The nonmonotonous $|\Delta(z)|$ inside the strip stems from the competition of an increase of pair breaking from the normalized flux approaching a half-integer (which suppresses $|\Delta|$) and the decrease of pair breaking from the increase in radius (which enhances $|\Delta|$).

When $|\delta n| > 1$, solitons in tapered tubes adopt arrangements that are different from the equal-z, necklacelike configurations of shorter bottlenecks [Fig. 2(g)]. In Figs. 3(c)–3(h) we show a series of long, tapered, nondestructive tubes of increasing R_+, with the number of solitons increasing from $\delta n = 2$ to $\delta n = 4$. Solitons maximize their relative distance by distributing along the tapered section to positions close to fluxoid jumps (vertical dashed lines), with strong variations of their φ position. The soliton configuration is no longer symmetric as in short bottlenecks, due to the frustration caused by a repulsion between multiple solitons, a situation reminiscent of frustrated antiferromagnets [97].

Discussion and outlook—We have demonstrated that bottlenecks in thin-walled superconducting tubes subjected to an axial magnetic field constitute boundaries between domains of potentially distinct topology. This happens as a result of a field-induced fluxoid mismatch δn. As a consequence, a number $|\delta n|$ of robust, topologically protected structures, dubbed fluxoid solitons, emerges within the bottleneck at equilibrium. Each of these solitons have a vanishing order parameter at their core and a width of the order of the superconducting coherence length.

Fluxoid solitons have a nonzero vorticity, apparent in the supercurrent density and local superconducting phase, and constitute a new type of superconducting vortex induced by the bottleneck geometry. They are reminiscent of Pearl vortices in thin superconductor films under a uniform out-of-plane magnetic field. However, there are important differences between the two. The flux through a Pearl vortex is usually (though not always [48,98]) asymptotically quantized due to screening beyond the Pearl length $2\lambda_L^2/t_s$ (where t_s is the thickness of the superconductor film and λ_L is the London penetration length). Instead, in the case of fluxoid solitons with fields parallel to the tube, the out-of-plane flux is restricted geometrically to the bottleneck region and it is not quantized even with magnetic screening. In fact, applying the Stokes theorem to a suitable contour, the total flux through the tube walls can be shown to be equal to $\Phi_+ - \Phi_-$. Another difference with respect to Pearl vortices is that fluxoid solitons are topological boundary modes. They appear at particular z locations determined by a different topological number at the left and right tube sections. Thus, they are only free to move in the azimuthal direction. In this regard, fluxoid solitons in tubes are more closely connected to solitonic solutions for the superconducting phase difference across planar Corbino Josephson junctions. There, a field-induced fluxoid mismatch in the junction leads to solitons or Josephson vortices pierced by a quantized flux (fluxons). The main difference with fluxoid solitons, apart from a lack of flux quantization, is that a tube bottleneck is not a weak-link Josephson junction but a continuous material of varying radius. This makes fluxoid solitons a two-dimensional, strong-coupling generalization of the one-dimensional solitons in Corbino Josephson junctions. Consequently, conventional approaches based on the sine-Gordon equation [55,99] are insufficient to characterize fluxoid solitons. The full quasiclassical Usadel theory employed here is instead required.

The supercurrents circulating around fluxoid solitons can be used to image them. The magnetic fields produced by these supercurrents are weak. We estimated them to be around ~0.1–1 mT in typical full-shell nanowires, forming a characteristic multipolar pattern around the bottleneck (see Supplemental Material [68]). Sensitive SQUID scanning microscopes can reach well below the micro-Tesla domain, so they could allow direct imaging of solitons [69,70]. A simpler, though less direct detection scheme, useful in sufficiently symmetric systems, would involve a sharp fall in the critical current as soon as one or more solitons enter the bottleneck. This stems from the same symmetry arguments behind the fluxoid-valve effect in full-shell Josephson junctions [40] (and, in general, in coaxially cylindrical and planar Corbino geometries [61–63,100–106]).

Fluxoid solitons are not static but can freely move inside the bottleneck along the azimuthal direction in tubes with cylindrical symmetry. Perturbations or defects that break this symmetry may lead to soliton pinning. Breaking cylindrical symmetry in a controlled way, on the other hand, could be used to drive solitons to specific angular positions. This can be done, for example, by misaligning the magnetic field relative to the cylinder axis (see Supplemental Material [68]). Solitons in nanowire bottlenecks could exploit this effect to detect field orientations, operating as the bubble in a spirit level. This is just one possibility afforded by the dynamical nature of solitons. Another striking phenomenon is the expected response to a phase bias (applied at a specific φ) across a fluxoid-mismatched bottleneck. Although ideally the phase bias should induce no supercurrent by virtue of the fluxoid-valve effect, it will, however, induce a global rotation of the solitons around the tube, by an angle proportional to δn and

the phase bias. Similarly, a finite voltage bias would translate into a time-dependent phase bias and, thus, a finite angular velocity of the solitons with an associated rotating ac Josephson microwave radiation [107,108]. A discussion of these and other effects of soliton dynamics in tubular bottlenecks is beyond the scope of this Letter and is left for future studies.

All simulations in this work were developed using the freely available Quantica.jl package [109].

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Data availability—The data that support the findings of this article are openly available [110].

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End Matter

Appendix: Evolution of solitons with system parameters—The behavior of the order parameter solution $\Delta(\boldsymbol{r})$ in a short bottleneck and, in particular, its dependence with system parameters, is even richer than Fig. 2 may suggest at first glance. Figures 4(a)–4(c) show in more detail how the $|\Delta|$ profile across the bottleneck changes as a function of T, $R_\pm$, and $\Phi_\pm$ parameters. This dependence generalizes the LP phenomenology of uniform tubes. As a brief summary of the LP effect, the value of $|\Delta|$ (also of the critical temperature and the spectral gap) in a uniform and thin-walled tube exhibits oscillations as a function of flux Φ, with maxima at integer Φ/Φ_0. Depending on the radius of the uniform tube, $|\Delta|$ may vanish (narrow tubes, destructive regime [96]) or reach finite minima (wider tubes, nondestructive regime) at half-integer Φ/Φ_0, which separate LP flux *lobes* with different n. Figures 4(d)–4(f) show the evolution of the asymptotic $|\Delta_\pm|$ far from the bottleneck for the simulations in Figs. 4(a)–4(c), which matches the above LP phenomenology.

Figures 4(a)–4(c) showcase the following qualitative behaviors of solitons. In Fig. 4(a) we see how an increasing temperature gradually "melts" the soliton, initially of radius $\sim\xi_0$ at low temperature, by making it wider until it dissolves away when one (or both) of the two sides of the bottleneck crosses the Φ-dependent LP critical temperature. Figures 4(a)–4(c) all show that the equilibrium position of a single soliton may shift away from the center of the bottleneck, moving toward the side with a weaker superconductivity. In Fig. 4(c) we encounter a peculiar situation for $\Phi_+/\Phi_0 \approx 1.6$–$2.2$ with $n_- = 0$ and $n_+ = 2$, so the bottleneck should in principle host two solitons, but in which the pairing on the n_- side of the bottleneck has collapsed (destructive regime). However, since the bottleneck is not sufficiently short, one of the two solitons survives, separated from the gapless portion of the tube by a short bottleneck region with finite Δ and local winding $n = 1$. This configuration is rather fine-tuned and fragile, not topologically protected like the solutions in the nondestructive regime with finite asymptotic Δ.

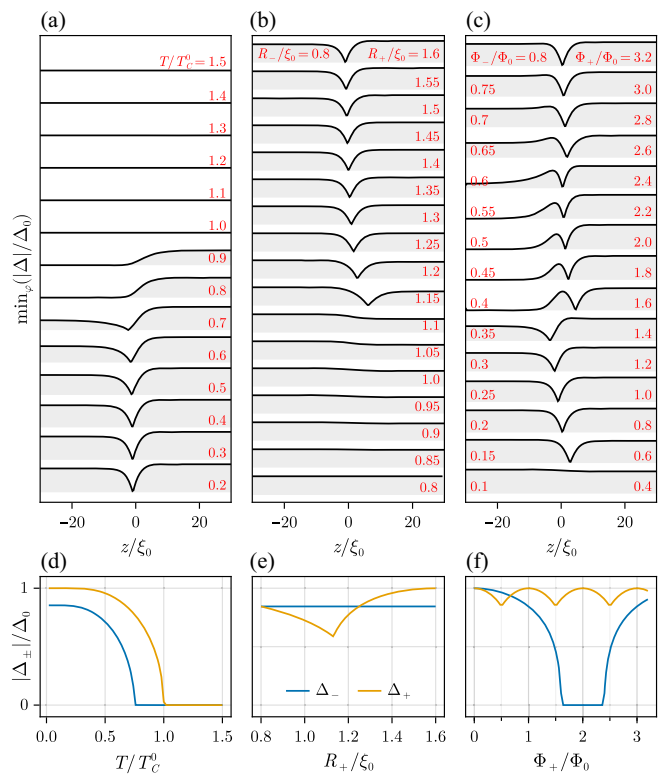

FIG. 4. Behavior of the soliton with system parameters. The soliton profile, $\min_\varphi(|\Delta|/\Delta_0)$, along z is plotted from bottom to top for increasing values of temperature (a), radius R_+ (b), and magnetic field (c). All other parameters are fixed in each case, and correspond to the bottleneck of Figs. 2(a) and 2(b). Panels (d)–(f) show the asymptotic $|\Delta_\pm|$ far from the bottleneck as a function of the same parameters.